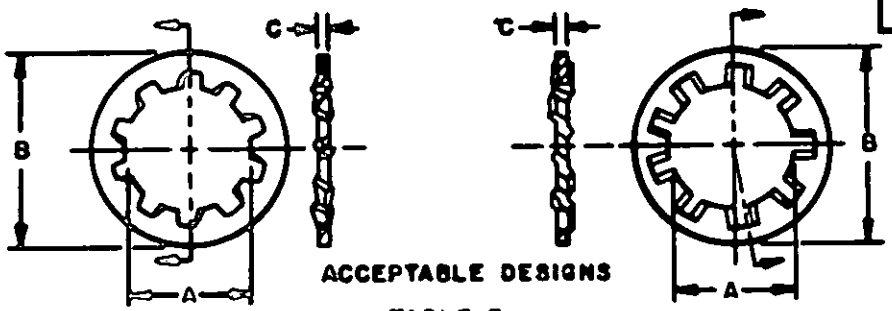
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5310



ACCEPTABLE DESIGNS

TABLE I

NOMINAL SIZE	A INSIDE DIAMETER		B OUTSIDE DIAMETER		C THICKNESS		STEEL			TIN-BRASS OR PHOSPHOR BRONZE
							CARBON	CORROSION RES		
								COMP 2 FF-W-100	300 SER.	
	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
#2 .086	.095	.089	.200	.175	.015	.010	35	69		103
#4 .112	.123	.115	.270	.255	.019	.015	36	70	153	104
#6 .138	(J) .150	.141	.295	.275	.021	.017	37	71	154	105
#8 .164	.176	.168	.340	.325	.023	.018	38	72	155	106
#10 .190	.204	.195	.381	.365	.025	.020	39	73		107
1/4 .250	.267	.255	.470	.460	.029	.023	40	74		108
5/16 .312	.333	.320	.610	.594	.034	.028	41	75		109
3/8 .375	.353	.333	.652	.670	.040	.032	42	76		110
7/16 .438	.464	.445	.780	.750	.046	.032	43	77		111
~~5/16 .500~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~			~~112~~
1/2 .500	.530	.513	.900	.867	.046	.037	44	78		113
~~9/16 .562~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~			~~114~~
9/16 .625	.663	.640	1.071	1.045	.050	.042	46	80		115
5/8 .750	.795	.769	1.245	1.220	.055	.047	47	81		116
7/8 .875	.927	.896	1.410	1.365	.060	.052	48	82		117
1 1.000	1.060	1.019	1.637	1.580	.067	.059	49	83		118
~~1-1/8 1.125~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~	~~.~~			~~119~~
1-1/4 1.250	1.325	1.275	1.975	1.921	.067	.059	51	85		120

TABLE II

NOMINAL SIZE	A INSIDE DIAMETER		B OUTSIDE DIAMETER		C THICKNESS		STEEL			TIN - BRASS OR PHOSPHOR BRONZE
							# CARBON	CORROSION RES		
								COMP 2 FF-W-100	300 SER	
	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1/4 .250	.267	.256	.410	.395	.020	.016	125	135	167	145
5/16 .312	.329	.319	.507	.493	.030	.020		136		
3/8 .375	.392	.383	.592	.580	.020	.016	131	141		151
7/16 .437	.448	.440	.592	.580			126	136		146
15/32 .468	.480	.472	.607	.593	.023	.013	127	137		147
9/16 .563	.580	.570	.690	.680	.023	.013	128	138		148
5/8 .625	.640	.630	.800	.790	.027	.017	129	139		149
3/4 .750	.785	.775	1.077	1.047	.027	.017	130	140		150
1 1.000	1.020	1.010	1.390	1.360	.028	.023				

*Washers shown in Table II are generally to be considered for use in mounting electronic components

#FOR CARBON STEEL WASHERS WITH BLACK FINISH REQUIREMENT ADD LETTER "B" AFTER DASH NUMBER. SEE REQUIREMENT 2.1 AND NOTE 4.

(J) denotes changes

APPROVED 4 MARCH 1954 REVISED (S) 6 DECEMBER 1977 (H) 22 AUGUST 1980 (J) 9 FEB 81

USER ACTIVITIES
ARMY - ME
NAVY - MC, OS YD

REVIEW ACTIVITIES
ARMY - AT, AV, ER, GL
DLA - IS, ES
NSA - NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

R.A. AR. OTHER CUST AS 99	TITLE: WASHER, LOCK, FLAT-INTERNAL TOOTH	MILITARY STANDARD MS 35333
PREPARING ACTIVITY: FF-W-100	SUPERSEDING: ANSOO IN PART, BEAX 1 AND BEAX 2	SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5310-1142

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Army - AT, AV, ER, GL
DLA-IS, ES
NSA-NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

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REQUIREMENTS:

1. MATERIAL: Carbon, Steel, Composition 1 in accordance with procurement specification.
Corrosion-resisting Steel, Composition 2 in accordance with procurement specification.
Corrosion-resisting Steel, 300 series, Rockwell C35-43 in accordance with Fed-STD-Do. 66.
Phosphor Bronze, Composition 3 in accordance with procurement specification.
Tin-brass, Composition 4 in accordance with procurement specification.

2. PROTECTIVE COATING: Carbon steel-Cadmium plated in accordance with QQ-P-416-Type II Class 2.
Corrosion-resisting steel - Passivate in accordance with procurement specification.

2.1 BLACK PROTECTIVE COATING: When required for Carbon steel, Composition 1 shall be black (Electrodeposited) chromium plating in accordance with MIL-C-14538.

NOTES:

1. DIMENSIONS: All dimensions are in inches.
2. The number of teeth shall be at the option of the manufacture.
3. Washer, dash numbers 121 and 112 only, have the alternate teeth twisted in a direction opposite to that of the other teeth.
4. PART NUMBER: The MS part number consists of the MS number, plus the dash number, plus the code letter, when applicable.

CODE LETTERS:

Add "B" as suffix to dash number for black finish per requirement 2.1. Example: MS 15333-1318 Washer, Carbon Steel, Black Finish.

5. For design feature purposes, this standard takes precedence over Procurement documents referenced herein.
6. Referenced documents shall be of the issue in effect on date of invitations for bids or request for proposal, except that referenced adopted industry documents shall give the date of the issue adopted.

LINED THROUGH NUMBERS ARE INACTIVE FOR NEW DESIGN AFTER DEC. 1977

INTERCHANGEABILITY TABLE

The following dash numbers of MS35333 AND AN936 are cancelled after 5 Dec 1963. The cancelled washers cannot always replace the new washers and should be used until existing stocks are depleted. Use only the new washers for design and engineering. Replacement shall be in accordance with this table.

TABLE III

PART NUMBERS											
CANCELLED		NEW	CANCELLED		NEW	CANCELLED		NEW	CANCELLED		NEW
AN936	MS35333	MS35333	AN936	MS35333	MS35333	AN936	MS35333	MS35333	AN936	MS35333	MS35333
A2	1	35	--	57	40	A1016	12	46	--	86	103
--	18	35	AA16A	--	**	--	29	46	A48	87	104
--	52	35	A516	7	41	--	63	46	A68	88	105
A3	1	**	--	26	41	A1216	13	47	A88	89	106
A4	2	36	--	53	41	--	30	47	A108	90	107
--	19	36	A616	0	42	--	63	47	AA168	91	108
--	53	36	--	25	42	A1416	14	48	A5168	92	109
A6	3	37	--	59	42	--	31	48	A6168	93	110
--	28	37	A716	9	42	--	65	48	A7168	94	111
--	54	37	--	26	43	A1616	15	49	A71685	--	112
A8	4	38	--	60	43		32	49		95	113
--	21	38	A7163	--	121		65	49		96	114
--	55	38	A816	10	44		16	50		97	115
A10	5	39	--	27	44		33	50		98	116
--	22	39	--	61	44		67	50		99	117
--	56	39	A916	11	45		17	51		100	118
AA16	6	40	--	28	45		35	51		101	119
--	23	40	--	62	45		63	51		102	120

**No replacement

APPROVED 4 MARCH 1954 REVISED (3) FOR CHANGES SEE SHEET 1

P.A.AR. OTHER CUST AS 99	TITLE: WASHER, LOCK, FLAT—INTERNAL TOOTH	MILITARY STANDARD MS 35333
PROCUREMENT SPECIFICATION FF-W-100	SUPERSEDES: AN936 IN PART, DEAN 1 AND DEAN 2	SHEET 2 OF 2

DD FORM 1 SEP 67 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5310-1142

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